GEOPHYSICAL INVESTIGATION OF AQUIFER CHARACTERISTICS IN THE UBIAJA AREA, ANAMBRA BASIN, NIGERIA

A Project Thesis Report

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Abstract

Groundwater is a critical freshwater resource for human consumption and agriculture, especially in regions with limited surface water. Geophysical methods like Vertical Electrical Sounding (VES) are used to locate and characterize subsurface aquifers by measuring the electrical resistivity of diferent geological layers(Lithology). In high elevation areas underlain by complex subsurface Lithology, identifying productive aquifers is very dificult, leading to a high rate of unsuccessful and costly borehole drilling. Here we use the VES method to determine the aquifer characteristics of the Ubiaja area, which is part of the Anambra Basin in Nigeria. We show that the aquifer is predominantly located in sand layers at depths between 242 and 293 meters with resistivity values ranging from 216.1 to 1885.8 Ωm. Our analysis further reveals that the southeastern part of the study area (VES 11 and 13) has the highest groundwater potential with superior transmissivity values up to 162,443 m^2/day and longitudinal conductance up to 0.1548 Ω^{-1}, contrary to the assumption that the entire area has uniformly poor aquifer potential. This study provides a definitive, spatially-explicit map of groundwater resources in Ubiaja, demonstrating that detailed geophysical investigation is indispensable for effective and sustainable groundwater development in geologically complex terrains.

Statement

Access to clean groundwater is a fundamental challenge in many rural communities. In the Ubiaja area of Nigeria, this challenge is more intense by difficult geological conditions, leading to widespread failed attempts to locate water through drilling. This research directly addresses this problem by using a geophysical technique to visualize underground(subsurface) and create a detailed map of the aquifer system. We identified specific locations with high water yielding potential, providing a clear scientific blueprint for where to successfully drill community boreholes. Our findings translate complex subsurface data into a practical guide for water resource development, which can help secure a sustainable water supply for the local population and serve as a model for similar regions in Africa facing water scarcity.

CHAPTER ONE: INTRODUCTION

Groundwater is water that is found in subsurface cracks and contained in pore spaces of sands and rocks. It is stored and moves slowly through geologic Formations of sand and rocks, called aquifers (Okeke et al, 2011). Groundwater resources of any area is controlled largely by geology, climate (Okeke et al, 2011), and is constantly recharged by groundwater infiltration, and surface waters. However, the quest for potable water makes for an increased drilling activities into the subsurface in search for groundwater, but the Ubiaja area does not give room for that, owing to the higher elevation, depth to water level being several metres into the subsurface, and bedrock of the area. Several unsuccessful borehole drillings confirms difficult it is to locate the aquifer in this area.

Hence vertical electrical sounding (VES) method is one of most useful techniques for geophysical investigation of aquifer characteristics in an area such as Ubiaja. Also if the area is characterized with more of impermeable rocks such as basement rocks with no cracks, the ease of fluid flowage becomes difficult and transmission of water from one area to another in the subsurface becomes a major problem.



Figure 1: Location map and elevation map of study area (Edo State where Ubiaja is located)

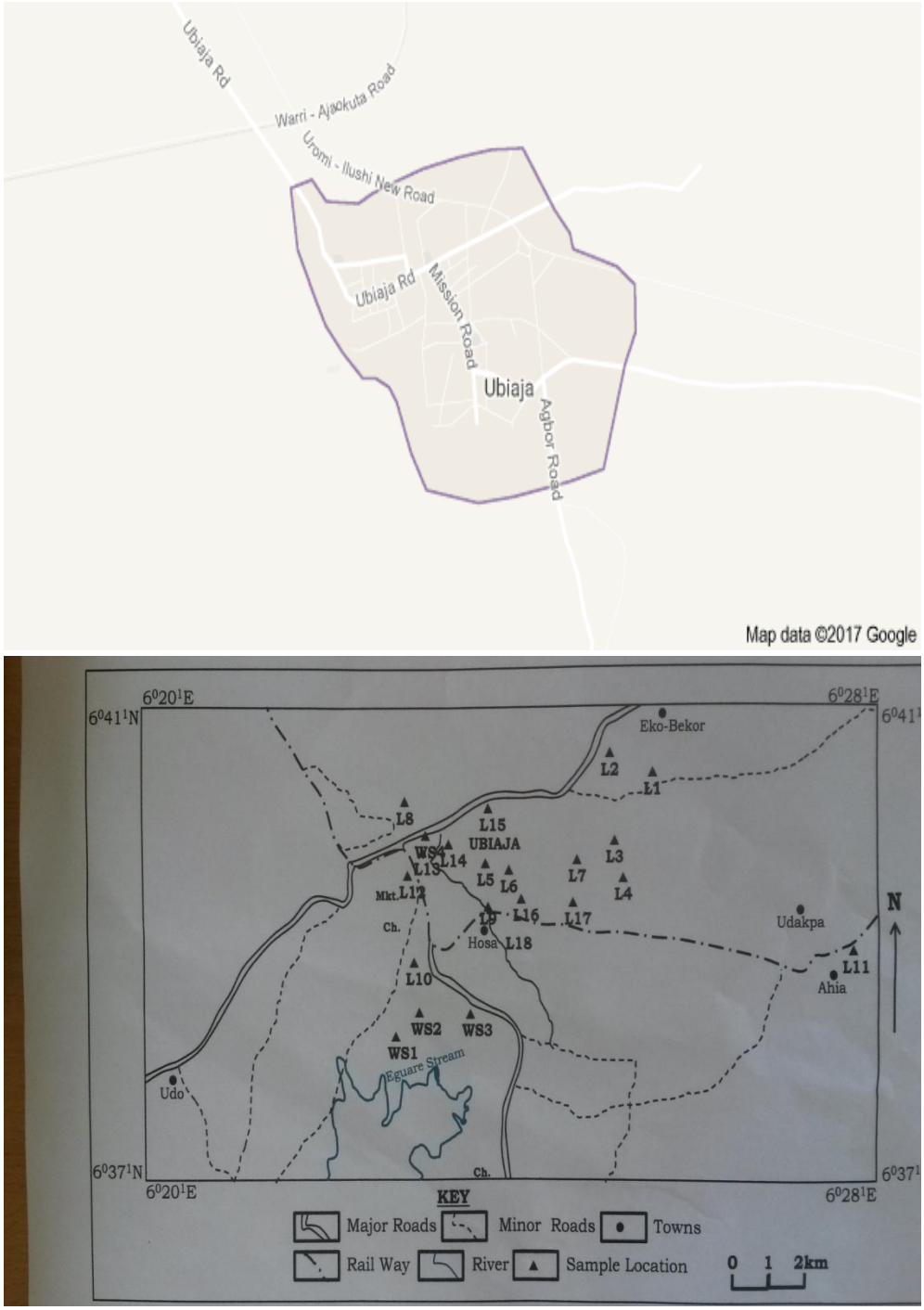


Figure 2: Detailed map of the study area in Edo State showing sample locations

Therefore we can speculate that Ubiaja is an area faced with high problem of aquifer location owing to high elevation and little or no geophysical knowledge of the subsurface. Hence the vertical electrical sounding (VES) method can be employed, to provide aquifer characteristics such as number of layers, layer thickness, depth to water level, hydraulic conductance and transmissivity. Infact this method has become very needful for the Ubiaja locals. The vertical electrical sounding method allows for the obtainment of quantitative results, by using

controlled source of specific dimensions (Odong, 2013). According to Okwueze (1996), depths of aquifers differ from place to place because of variation in geostructural occurrence.

The overall objective of this study is to determine aquifer characteristics of the study area an provide an in-depth knowledge of the subsurface through geophysical sounding method (VES). This method will derive the resistivity of the subsurface geologic boundaries from conductivities and use same to determine geoelectric sections (Odong, 2013).

Literature Review

Although not much work has been carried out on Ubiaja using geophysical method, but available information shows the area to be characterized by higher elevation, and this is further confirmed by Okeke et al (2011), who stated that the entire Ishan destrict is characterized by higher topography.

Furthermore, at southern Obubra, southeastern Nigeria, an area characterized by higher topography as Ubiaja. Odong (2013), was able to use geophysical method to investigate the aquifer characteristics of Obubra. Odong (2013), stated that aquifer at Obubra can be up to 140m thick, and can be located in the Western part. He also added that if thick saturated shale is found in an area, this can be misleading to be prolific aquifer units to most drillers, and that to develop groundwater at the Obubra area, bore constructions must be preceded by a detailed geophysical investigation (VES), boreholes must be logged to determine exact screen positions.

In line with borehole logging as suggested by Odong (2013), Akpoborie and Efobo (2014) stated that through recovering of drill cutting and subsequent grain size analysis, much can be known about grain size and hydraulic characters of the aquifer or subsurface in an area.

Also in using VES method for geophysical investigation, Akpoborie and Kizito (2012) noted that areas other than the top soil having low apparent resistivity values may imply aquifer layers in the subsurface.

In trying to predict aquifer protective capacity of the Agbani Sandstone, using Dar-Zarrouk parameters, Austin and Gabriel (2015) were able to use vertical electrical sounding method to estimate apparent resistivity method, data were plotted against electrode spacing and respective aquifer layer resistivity, thickness, depth obtained. From the afore mentioned Austin and Gabriel were able to estimate Dar-Zarrouk parameters; transmissivity and hydraulic conductance, and hence predicted aquifer protective capacity zones to be moderat, good and poor.

Various researchers that were able to use Dar-Zarrouk parameters to determine aquifer parameters include; Onuoha and Mazi (1988). They used Dar-Zarrouk parameters to estimate transmissivity of Ajali Sandstone aquifers in Southeastern Nigeria. Ezeh (2011) also appliedthe concept of Dar-Zarrouk parameters for aquifer hydraulic properties.

With great success from world-wide, various attempts have been made to determine transmissivity using empirical and semi-emperical relationships to, some of these researchers include Kelly (1985); Huntley (1986); Urish (1987); Chen et al (2001); Singh (2005).

Aim and Objectives

The major aim of the study is to determine aquifer parameters from vertical electrical sounding (VES). The objectives are:

- Longitudinal conductance
- Transmissivity
- Aquifer thickness
- Depth

Physiographic Setting

Climate and Vegetation: Ubiaja is a tropical climate area, with distinct rainy season around April to October and dry season around November to March. The study area is a higher topographical area, characterized by tropical vegetation such as the Guinea forest.

Drainage and Accessibility: The study area is characterized by underground drainage, and is accessible through major and minor roads linking it to neighboring communities.

Settlement and Land Use: According to the Nigerian Meteorological agency (2007), the study area is semi-urban area, average annual rainfall is 1704mm. the uppermost deposit of Ajali sandstone is reddish and fertile. Consequently the community is agrarian.

CHAPTER TWO: TECTONIC EVOLUTION AND STRATIGRAPHY

2.0 Tectonic Evolution of the Anambra Basin

Following the Mid-Santonian tectonism and magmatism, depositional axis shifted westward resulting in the subsidence of the Anambra Basin of which Ubiaja is part of. The Anambra Basin is taken as part of the lower Benue Trough that contains the post deformational sediments of Campanian to Maastrichtian to Eocene. It is logical to include the Anambra Basin in the Benue Trough being a related structure that developed after the compressional stage (Akande and Erdtmann, 1998). Thus post deformation and sedimentation in the Lower Benue Trough constitutes the Anambra Basin.

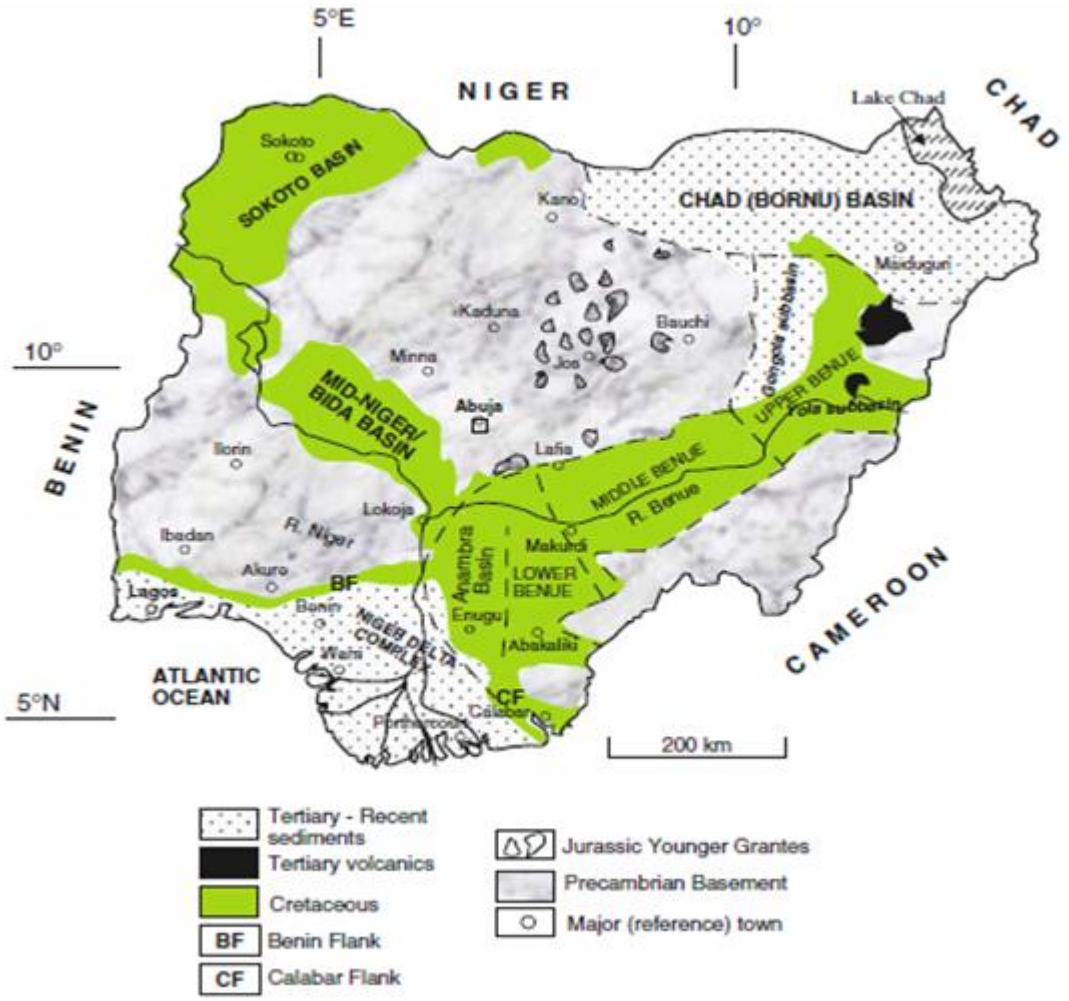


Figure 3 : Geologic map of Nigeria showing the Anambra Basin (Obaje, 2009)

2.1 Sedimentation and Stratigraphy

Sedimentation in the Anambra Basin thus commenced with the Campanian to Maastrichtian marine and paralic shales of the Enugu and Nkporo Formations, overlain by the coal measures of the Mamu Formation. The fluviodeltaic sandstones of the Ajali and Owelli Formations lie on

the Mamu Formation and constitutes its lateral equivalent in most places. In Paleocene the marine shales of the Imo and Nsukka Formation where deposited and overlain by the tidal Nanka Sandstone of Eocene age.

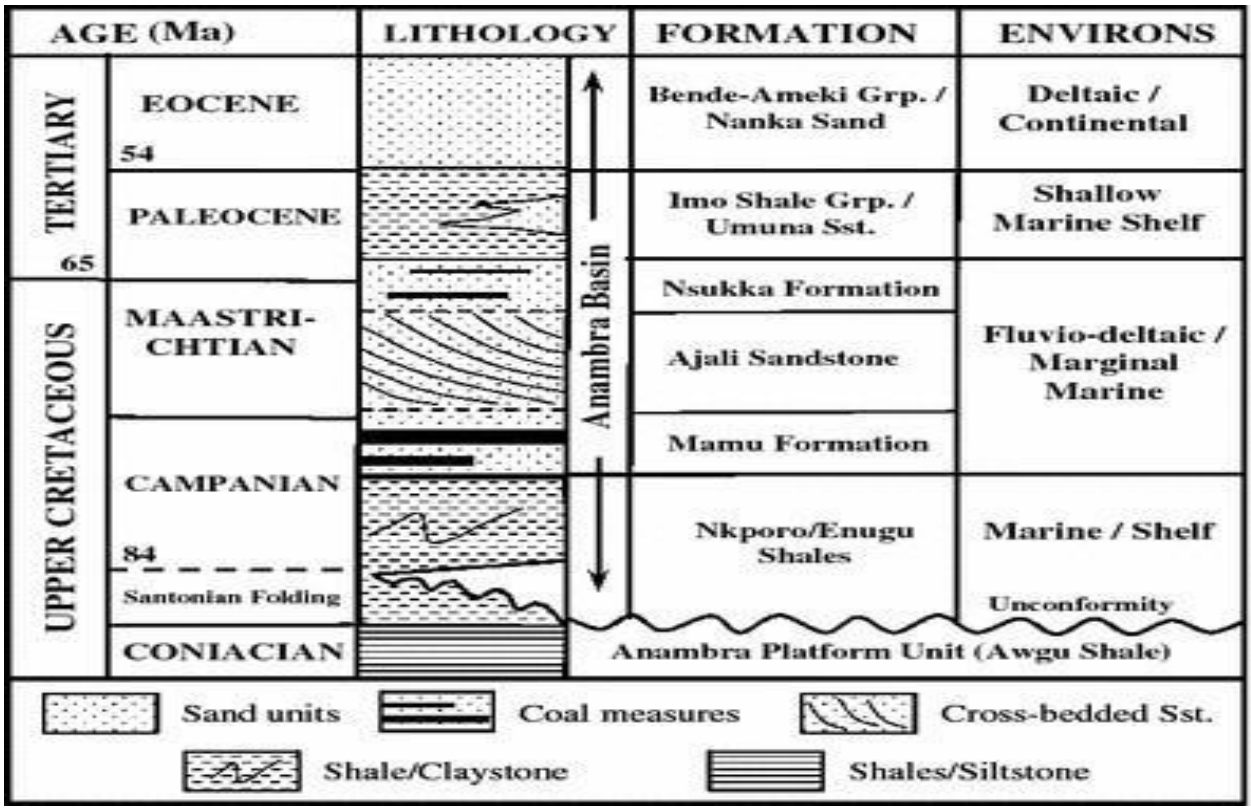


Figure 4 : Stratigraphic profile and Age/Lithology chart of the Anambra Basin

CHAPTER THREE: METHODOLOGY

The ABEM SAS 1000 terrameter was used to measure the apparent resistance that the ground offers to the flow of electric current through it. The terrameter automatically records voltage and current. The electrode arrangement used for acquiring data is the schlumberger arrangement of electrodes. Soundings were performed at eighteen (18) locations with maximum current electrode spacing ranging from 1m to 250m. Observed field data were plotted manually and using the WinResist software, thus producing the number of layers, with layer apparent resistivity, layer thickness and depth. Hydraulic conductance and transmissivity were derived using the following formula:

$K = h / R$
$T = Kh$

Where h=thickness, R=apparent resistivity, K=Hydraulic conductance, T=Transmissivity. K and T were named as Dar-Zarrouk parameters by Maillet (1947). Using the surfer14 software, the aquifer resistivity and transmissivity map were produced and also the water table map was produced.

3.1 Field Operational Practice



Figure 5: Field setup photo of schlumberger array data acquisition team at Ubiaja

Equipment used: Terrameter, Battery, Wire, Umbrella, Hammer, Pegs.

Field procedure involves increasing the two outer current electrodes as shown in the array diagrams with a fixed inner potential electrode spacing. Field measurement for current electrode spacing ranged from 1m to 250m.

Theory of Schlumberger Arrangement

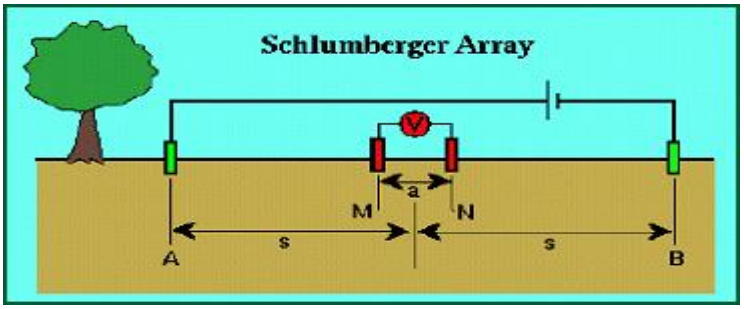


Figure 6: Schematic diagram of a typical Schlumberger Arrangement configuration

Generally four electrodes are needed for this set-up: two outer current electrodes AB and two inner potential electrodes MN. Distance between the potential electrodes MN must never exceed 2/5 of AB/2. Where AB are the two outer current electrode. With the above, we therefore have MN ≤ AB/5.

The mathematical derivations define the geometric factor K such that apparent resistivity r = KV/I = KR. Curves are classified into H, A, Q, or R type variants depending on the stratification sequence.

CHAPTER FOUR: RESULTS AND DISCUSSIONS

A total of eighteen (18) locations were sounded. Field data curves were modeled using specialized software to identify subsurface parameters.

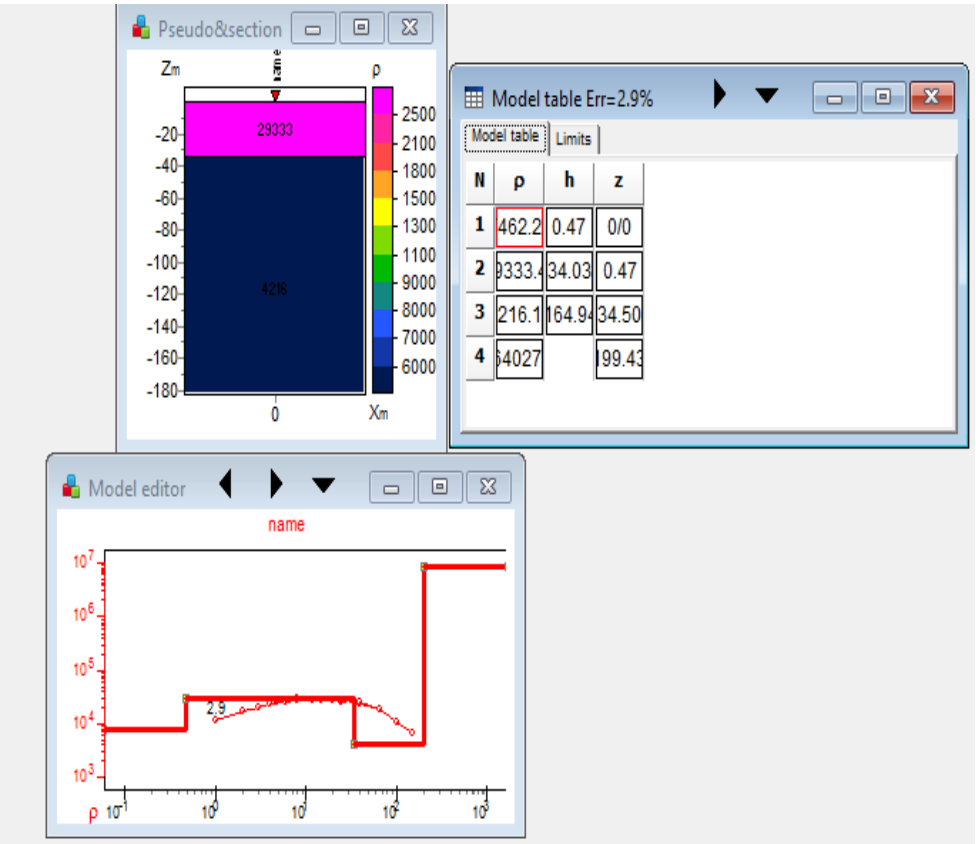


Figure 7a: ZONDIP1D Inverted geoelectric sounding curves (VES 1)

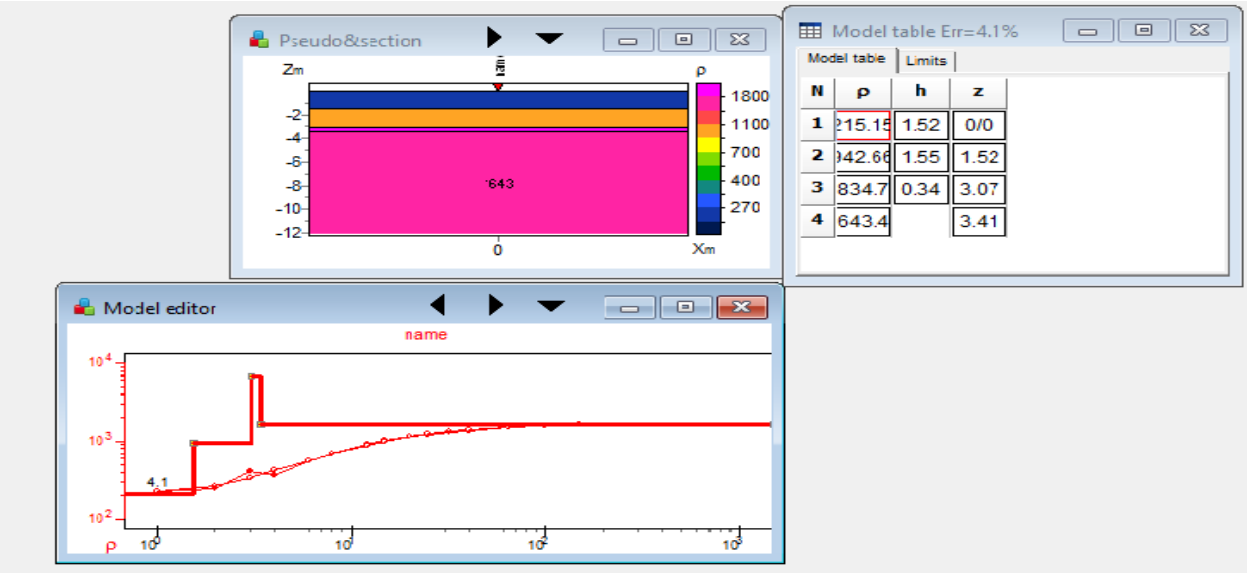


Figure 7b : ZONDIP1D Inverted geoelectric sounding curves (VES 6)

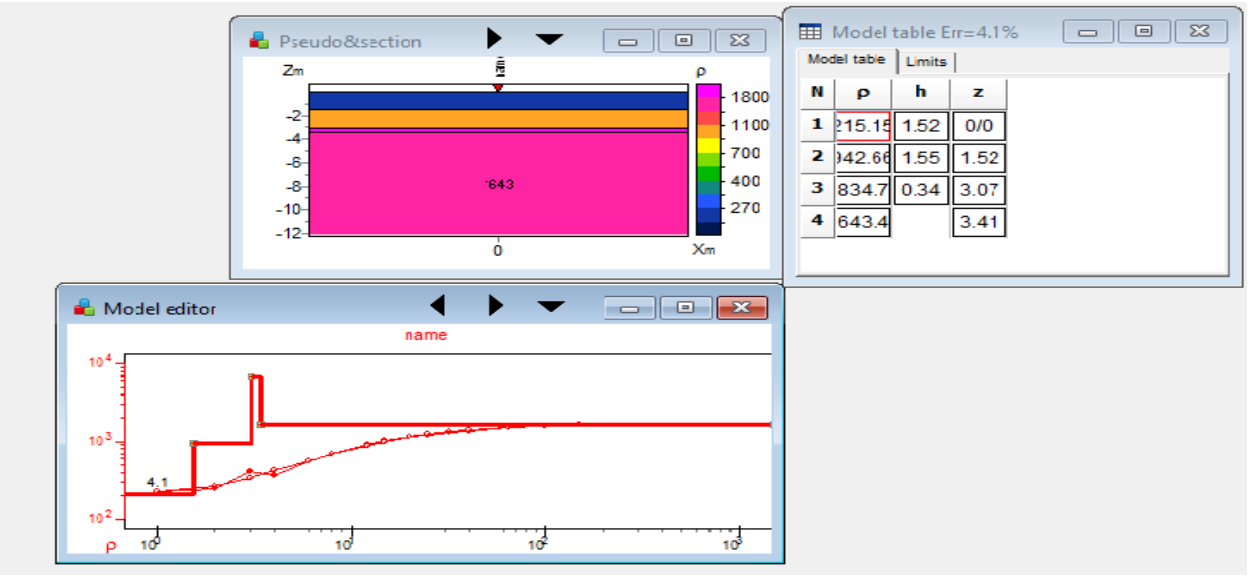


Figure 7c : ZONDIP1D Inverted geoelectric sounding curves (VES 10)

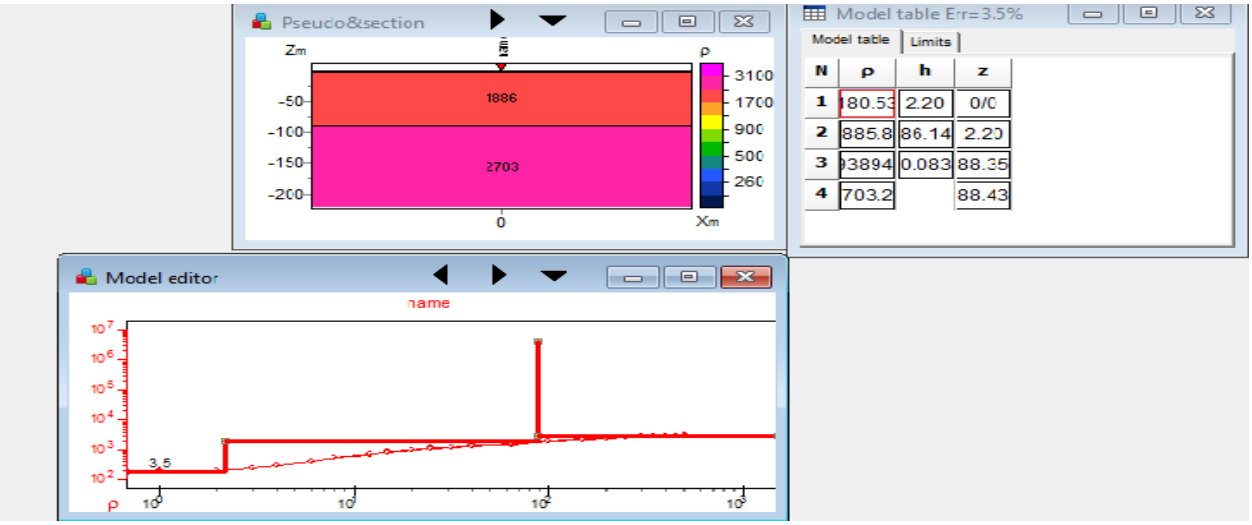


Figure 7d : ZONDIP1D Inverted geoelectric sounding curves (VES 11)

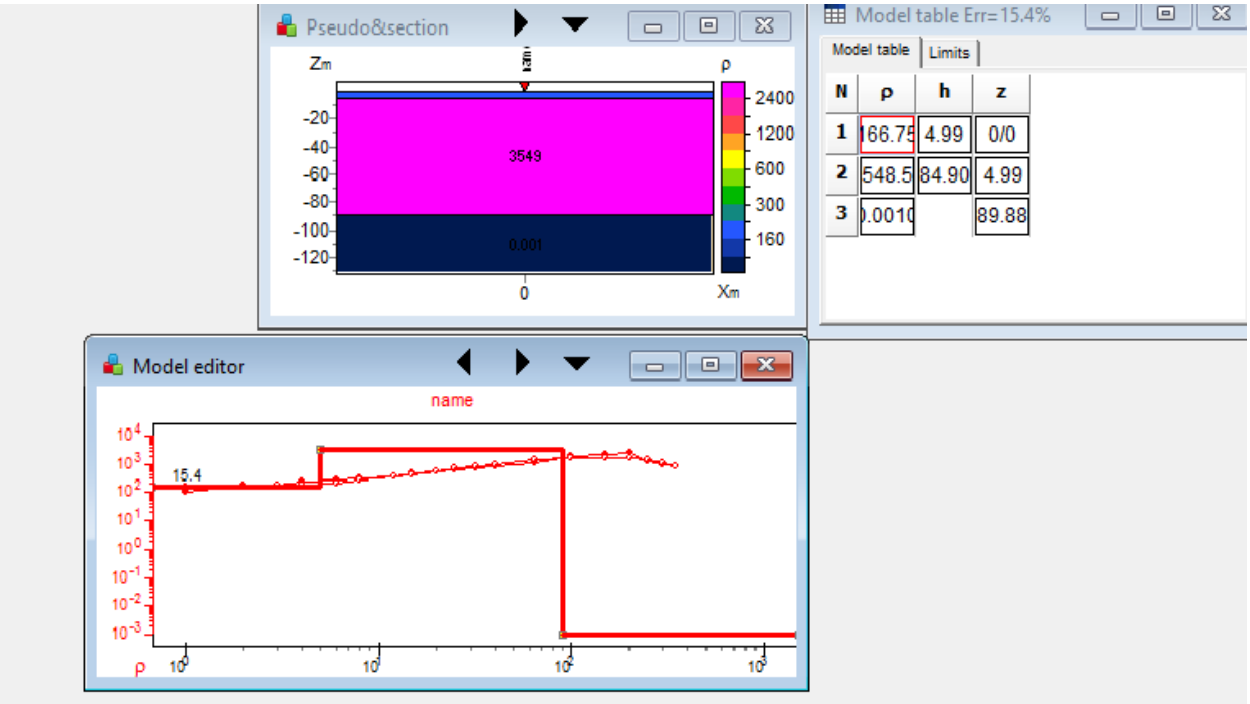


Figure 7e : ZONDIP1D Inverted geoelectric sounding curves (VES 13)

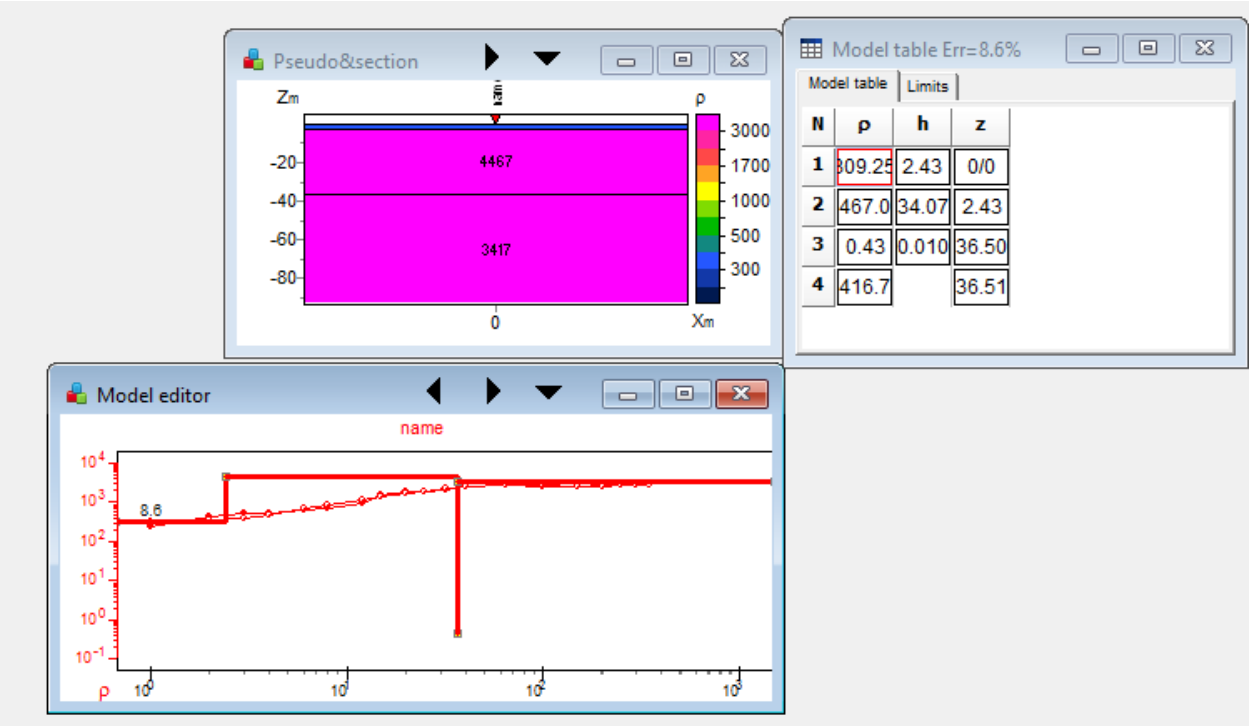


Figure 7f : ZONDIP1D Inverted geoelectric sounding curves (VES 16)

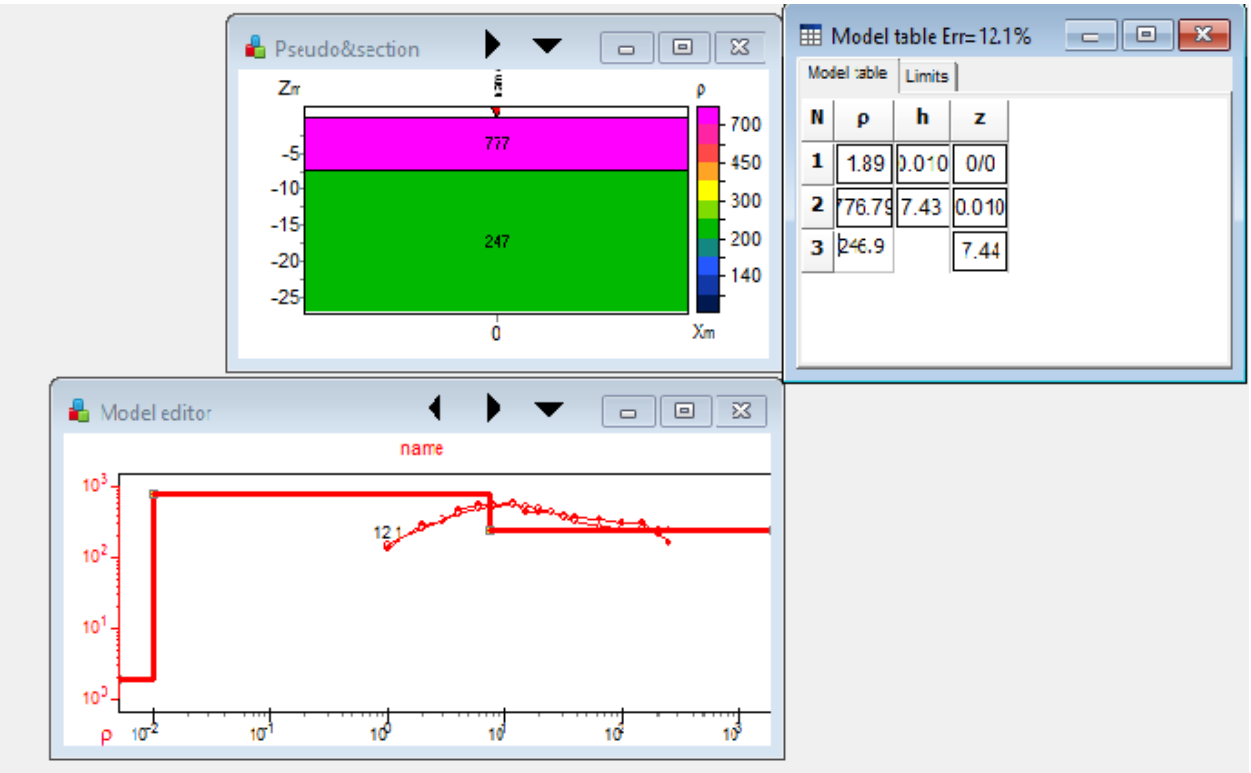


Figure 7g: ZONDIP1D Inverted geoelectric sounding curves (VES 18)

Table 1: Aquifer Parameters from Data Interpretations

VES No	Layer	Resistivity (Ωm)	Depth (m)	S (Ω^{-1})
1	3	216.1	242.5	0.0784
6	2	1215.26	287.24	0.0021
10	2	1777.34	280.41	0.0011
11	2	1885.8	248.8	0.0457
13	2	548.5	293.01	0.1548
16	2	467	274.57	0.0730
18	2	776.79	267.99	0.0096

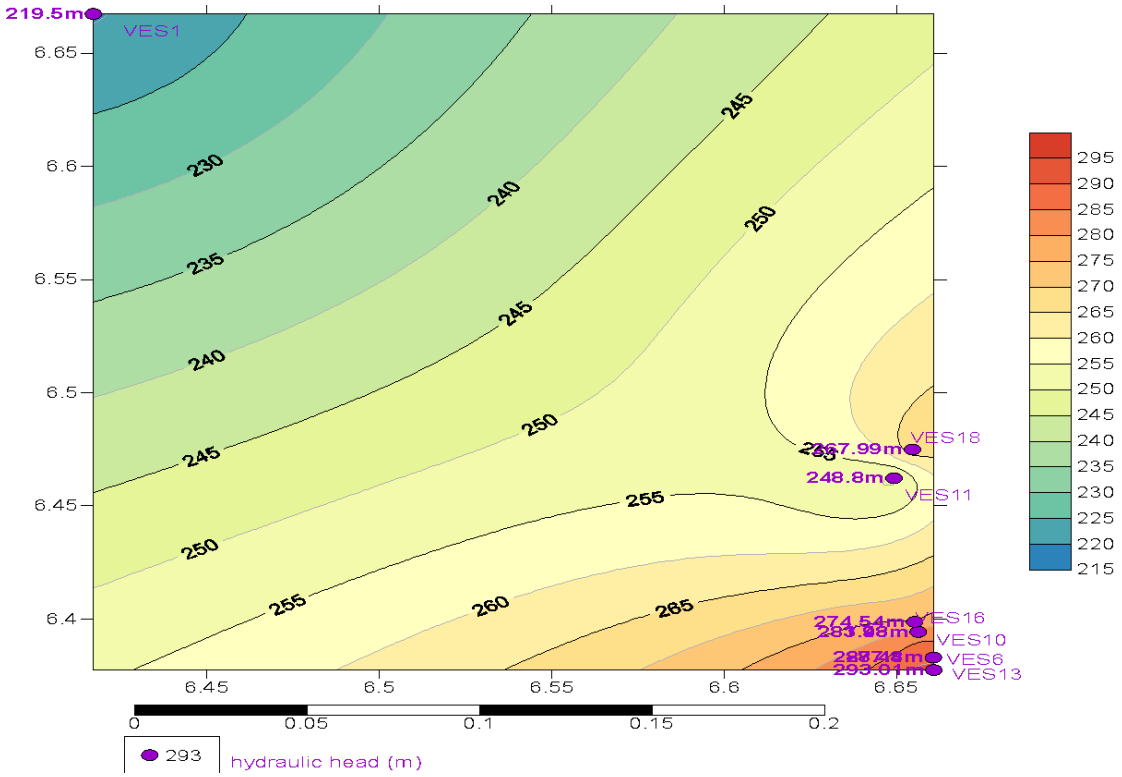
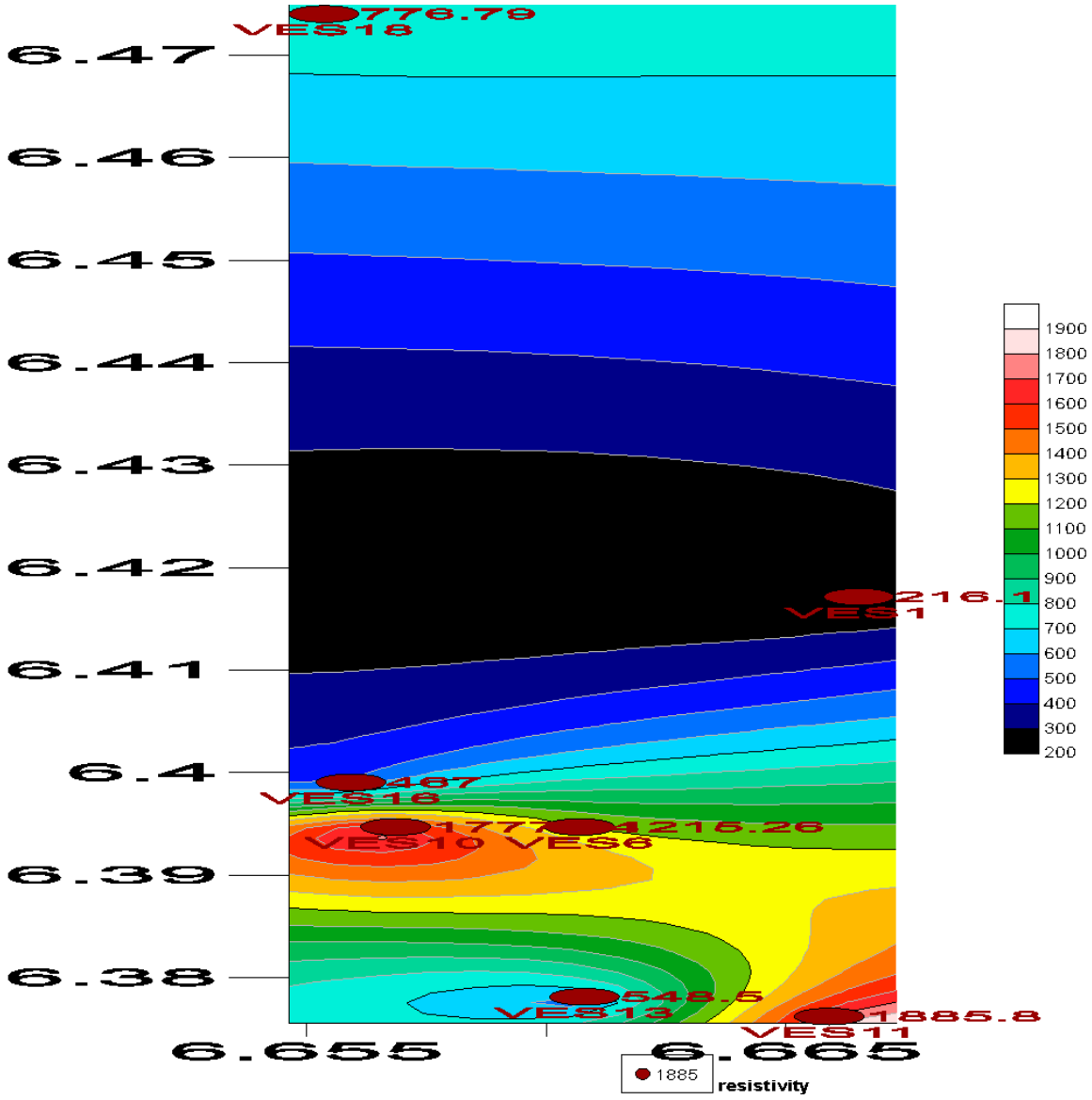


Figure 8: Water table contour map of the study area


776.79
VES18
6.47
6.46
6.45
6.44
6.43
6.42
216.1
VES1
6.41
6.4
467
VES16
1777
1215.26
VES10
VES6
6.39
6.38
548.5
VES13
1885.8
VES11
6.655
6.665
1885
resistivity
1900
1800
1700
1600
1500
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200

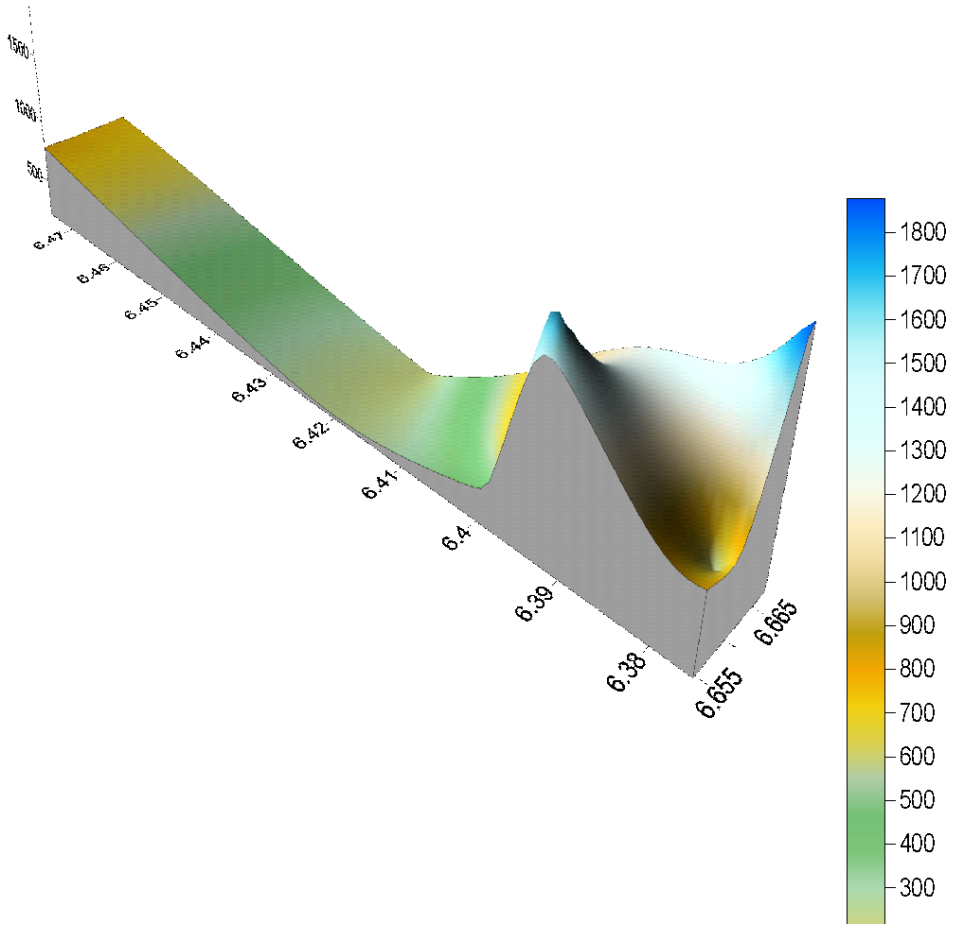
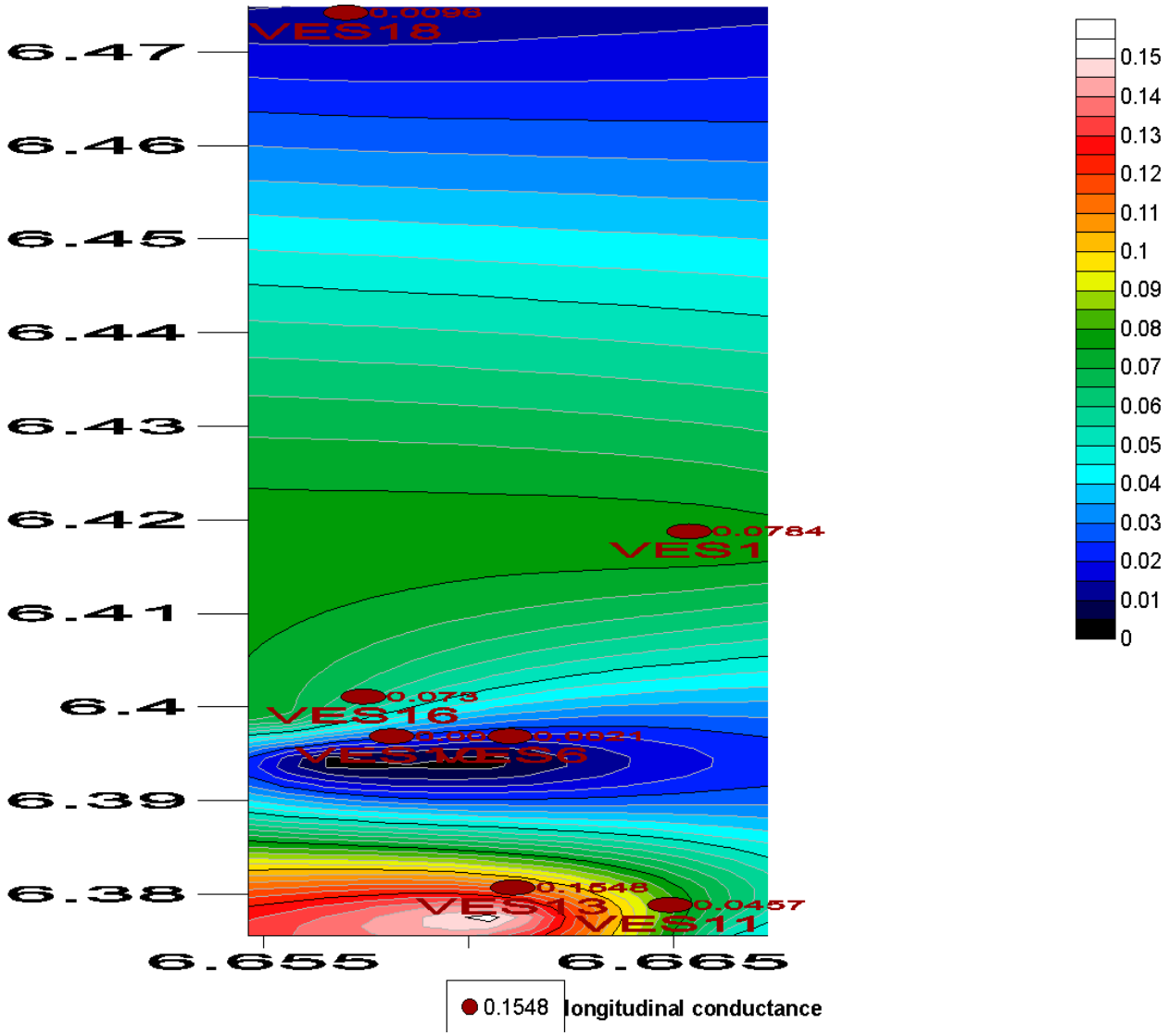


Figure 9 : Subsurface resistivity distribution maps

Table 2: Coordinates and calculated hydraulic heads

Latitude	Longitude	Hydraulic Head (m)	VES Location
6.4171169	6.6673109	219.5	VES 1
6.6606997	6.3830057	287.8	VES 6
6.6606997	6.3830057	287.44	VES 6
6.656333	6.394306	283.48	VES 10
6.656333	6.394306	281.93	VES 10
6.649250	6.462222	248.8	VES 11
6.660806	6.377472	293.01	VES 13
6.655278	6.398722	274.54	VES 16
6.654639	6.474861	267.99	VES 18


6.47
6.46
6.45
6.44
6.43
6.42
6.41
6.4
6.39
6.38
6.655
6.665
0.0098
VES18
0.0784
VES1
0.073
VES16
0.0021
VES15
VES6
0.1548
VES13
0.0457
VES11
0.15
0.14
0.13
0.12
0.11
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0.1548
longitudinal conductance

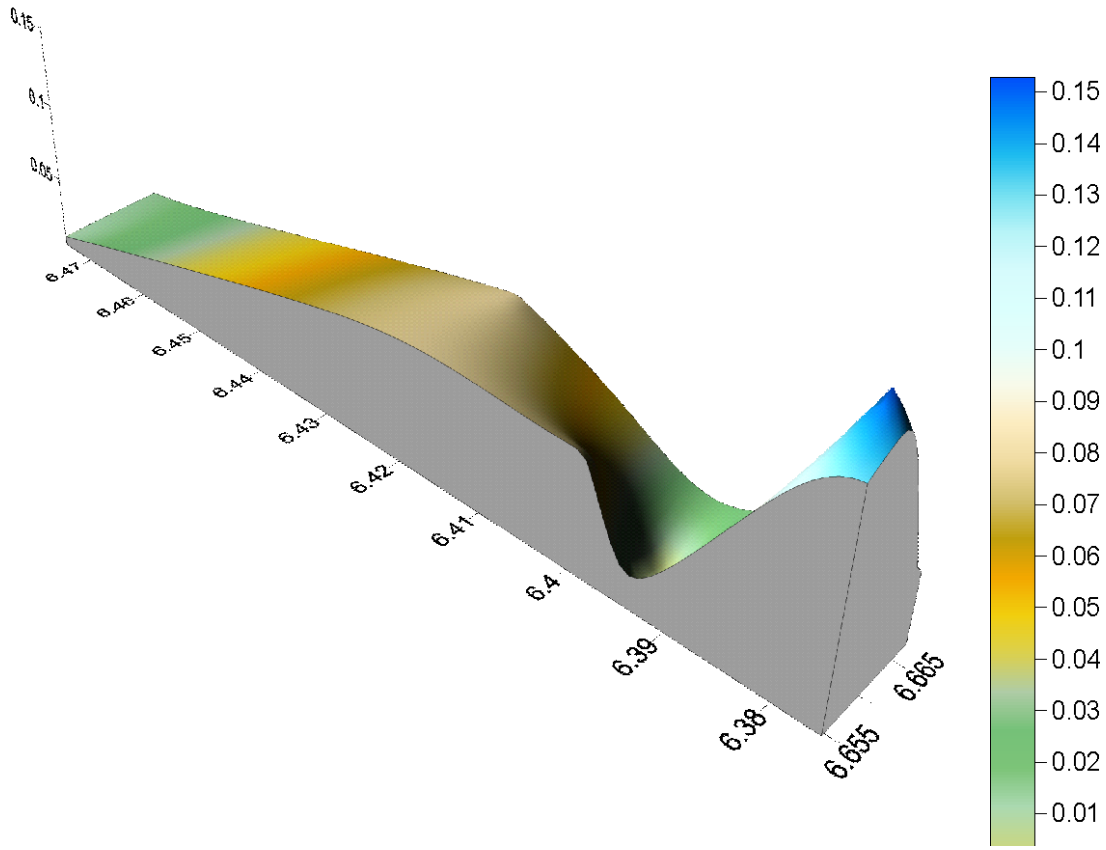
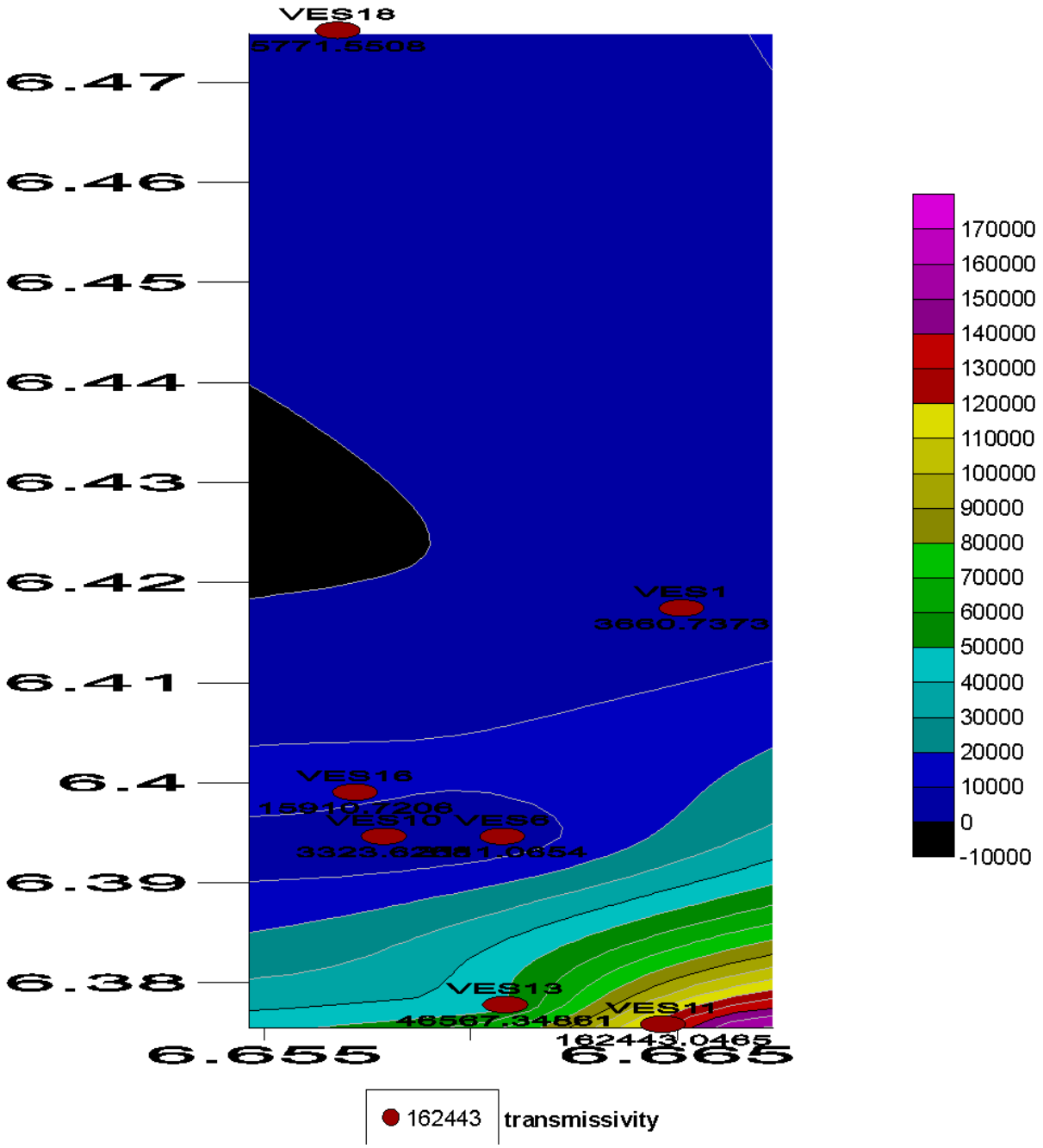


Figure 10 : 2D and 3D longitudinal conductance distribution maps


VES18
5771.5508
6.47
6.46
6.45
6.44
6.43
6.42
6.41
6.4
6.39
6.38
VES1
3660.7373
VES16
VES10
VES6
3323.62
1.0654
VES13
40567.34861
VES11
162443.0465
6.655
6.665
170000
160000
150000
140000
130000
120000
110000
100000
90000
80000
70000
60000
50000
40000
30000
20000
10000
0
-10000
162443
transmissivity

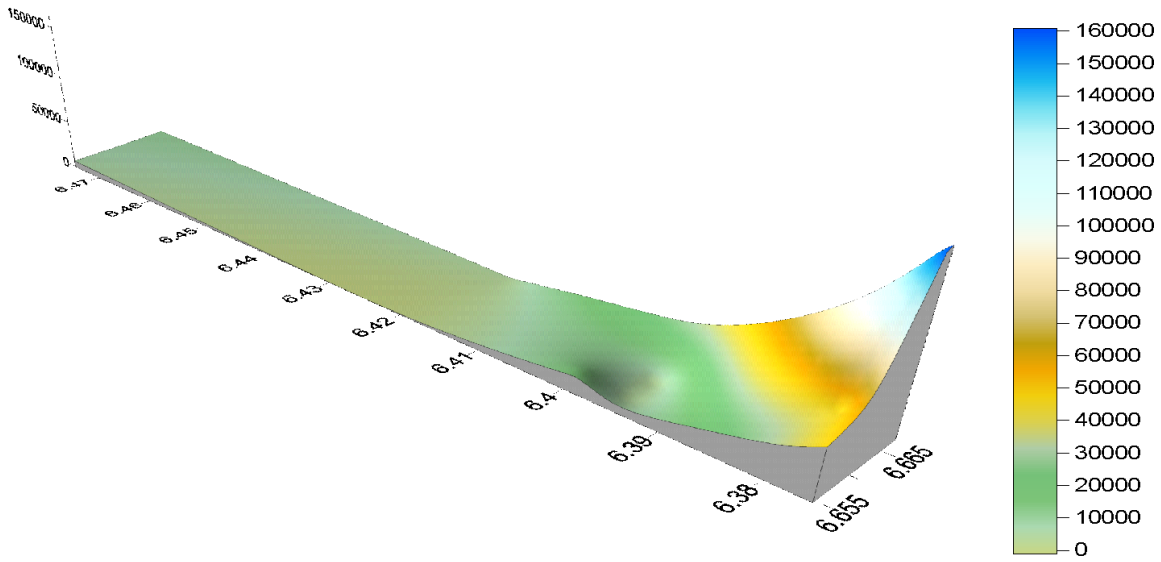


Figure 11 : 2D and 3D aquifer transmissivity contour maps

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

Conclusion

Conclusively, from this study, we can say that the aquifer zone of the Ubiaja area lies in the sand formations. The overall resistivity values range from 216.1 to 1885.8 Ωm, and longitudinal conductance ranges from 0.0784 Ω^{-1} to 0.1548 Ω^{-1}. The locations with the highest aquifer potential are VES 11 and VES 13 in the southeastern direction.

Recommendations

It is highly recommended that to have good knowledge of aquifer parameters which help show areas of good groundwater potential, geophysical investigation should be carried out prior to drilling. It is advised to drill industrial-scale communal boreholes in the southeastern sector to serve the entire Ubiaja community.

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